

Seenivasan Hariharan, Ph.D.

Quantum Computing & Computational Chemistry Researcher

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Profile

Application-driven quantum computing researcher and educator with over nine years of postdoctoral experience integrating computational chemistry, high-performance computing, and quantum algorithm design. My expertise spans DFT and molecular dynamics simulations on HPCs, development of chemistry-inspired quantum algorithms, and hands-on experimentation on superconducting quantum hardware with advanced error-mitigation and transpilation techniques. I thrive at the interface of science, technology, and communication—bridging researchers, engineers, and business stakeholders to translate quantum methods into practical impact. My recent work includes a perspective on quantum computing for catalysis, a qubit-routing algorithm for lattice simulations, and resource-aware methodologies supporting quantum chemistry applications. Recognized for mentoring, course development, and leadership in collaborative, multi-stakeholder research initiatives.

Technical Highlights

- **Quantum Hardware:** Executed variational quantum algorithms and noise-aware chemistry simulations, on IBM's 127-qubit *Eagle* and 156-qubit *Heron* processors (awarded \$165K compute grant); evaluated trapped-ion and neutral-atom platforms through QAL-SURF collaborations for education.
- **Qubit Routing Algorithm Development:** Co-developer of a lattice-based qubit routing algorithm for quantum simulation published in (*ACM TQC*, 2025) with an accompanying code in a GitHub repository.
- **Quantum Algorithm Development:** Co-developer of a fault tolerant quantum algorithm based on quantum random walks for analysing chemical reaction networks *Submitted to QIP 2026 and on arxiv: 2509.07890v1*.
- **Computational Chemistry & Simulation:** Over five years of experience in DFT and molecular dynamics simulations on HPC systems using **VASP**, **FHI-AIMS**, **OpenMOLCAS**, and **PySCF**, applied to surface chemistry and heterogeneous catalysis.
- **Teaching & Certification:** Co-designed and co-taught the *Near-Term Quantum Computing* course in the new MSc in Quantum Computer Science program at the University of Amsterdam, providing students with hands-on access to superconducting, trapped-ion, and neutral-atom quantum hardware. Awarded the Dutch University Teaching Qualification (**BKO**, 2025).
- **Collaborative R&D:** Lead researcher in the University of Amsterdam–Toyota Motor Europe–Vrije University Amsterdam project on quantum algorithms for modelling catalytic reactions; contributed to a TKI-funded software development grant (QC² 2.0, €50K).
- **Professional Recognition:** IBM Certified Associate Developer – Quantum Computation using Qiskit v0.2X (2023); awarded multiple IBM Quantum Challenge Advanced Badges (2022–2024).

Key Skills

Quantum Computing: Qiskit, PennyLane, Variational Algorithms, Error Mitigation, Generative Quantum Eigensolver (GQE)

Programming: Python (NumPy, SciPy, Pandas), Bash, FORTRAN, Git

Computational Chemistry: DFT, Ab-initio methods, Reaction Dynamics, Surface Chemistry, Phonons, Catalyst Design

Visualization & Data: Matplotlib, gnuplot, IRASPA, VESTA, Materials Studio

Professional Experience

Self-employed, Leiden, The Netherlands

Founder & Consultant | Jan 2026 – present

- Consulting on research projects related quantum computing for quantum chemistry and on creating modules for training programs in quantum computing.

University of Amsterdam, QuSoft & Quantum Application Lab (QAL), The Netherlands

Postdoctoral Researcher | Oct 2021 – Dec 2025

Advisors: Prof. Kareljan Schoutens, Prof. Lucas Visscher, Dr. Sachin Kinger

- Led application-driven quantum R&D for chemistry and materials, developing variational, multiscale, and network-based quantum algorithms with clear relevance to industrial catalysis use cases.
- Coordinated the theory pillar of the multidisciplinary roadmap “Spin Matters” (2025), translating fundamental limitations of DFT into a strategic case for quantum and beyond-DFT methods in industrially and societally relevant water-splitting catalysis.
- Implemented hybrid quantum–classical workflows for photocatalytic reaction modelling, contributing to externally visible outputs (perspective article, forthcoming manuscript) that position quantum methods for downstream adoption.
- Supervised integrated BSc/MSc projects on hardware-aware quantum algorithms, aligning student research with platform capabilities, emerging IP opportunities, and application-ready quantum workflows for chemistry and optimization.
- Proposed and co-developed a novel quantum algorithm applying quantum random walks to chemical reaction network analysis, opening new pathways for accelerated kinetics analysis and mechanism discovery beyond classical scaling limits, with clear potential for industrial process optimization.

Leiden University, The Netherlands

Postdoctoral Researcher | Dec 2018 – Sep 2021

Advisor: Dr. Jörg Meyer

- Performed high-accuracy DFT and dynamics simulations of CO adsorption and diffusion on alkali-halide and metal surfaces, earning the *CHAINS 2020 Best Poster Award*.
- Investigated vibrational energy dissipation and phonon lifetimes at metal surfaces, establishing key insights into energy-transfer mechanisms; supervised an MSc project that received the Unilever Research Prize.
- Guided undergraduate and master’s students on projects bridging theory and experiment in surface chemistry, experiences that later informed my quantum simulation work on lattice and spin models.

Texas Tech University, USA

Postdoctoral Research Associate | Mar 2017 – Nov 2018

Advisor: Prof. William L. Hase

- Conducted direct chemical-dynamics simulations of O₂ interactions with defective graphene, revealing new oxidation mechanisms and benchmarking reactivity at defect-rich carbon interfaces.
- Explored O₂ adsorption on defective graphene as a multi-reference problem, deepening understanding of reaction pathways that later guided my interest in multiconfigurational approaches.

Indian Institute of Science Education and Research (IISER) Kolkata, India

Ph.D. Researcher &

Postdoctoral Fellow | Aug 2011 – Mar 2017

Advisor: Prof. Ashwani K. Tiwari

- Performed quantum-mechanical and semiclassical studies of water and methane dissociation on nickel surfaces and nanoclusters, producing four first-author publications.
- Built a foundation in reaction dynamics, catalysis, and electronic-structure theory in bimetallic catalysis.

Education

Ph.D. Computational and Theoretical Chemistry, IISER Kolkata, India, CGPA 8.75, (2016)

Thesis: *Water and Methane Dissociation on Nickel Surfaces and Nanoclusters*

M.Sc. Chemistry, Bharathiar University, India – 86.06%, Department Topper

B.Sc. Chemistry, Bharathiar University, India – 84.20%, Department Topper

Selected Publications

Total publications: 23 (published as **Seenivasan Hariharan** and **H. Seenivasan**); Citations: 620; h-index: 14; i10-index: 17 (as of 1st Jan 2026).

1. **Hariharan, S.***, Zur, S.*, Kinge, S., Schoutens, K., Visscher, L., and Jeffrey, S. *Chemical Reaction Networks and Quantum Random Walks* (2025). *Both authors contributed equally. arXiv:2509.07890v1.
2. van der Minne, E., Vensaus, P., Ratovskii, V., **Hariharan, S.**, et al. *Spin Matters: A Multidisciplinary Roadmap to Understanding Spin Effects in Oxygen Evolution Reaction during Water Electrolysis*. **Advanced Energy Materials** (2025): e03556.
3. **Hariharan, S.**, Kinge, S., and Visscher, L. *Modelling Heterogeneous Catalysis using Quantum Computers: An Academic and Industry Perspective*. **Journal of Chemical Information and Modeling**, 65(2), 472–511 (2025).
4. Kattemölle, J., and **Hariharan, S.** *Line-Graph Qubit Routing*. **ACM Transactions on Quantum Computing**, 6(22), 1–18 (2025).
5. **Hariharan, S.**, Majumder, M., Edel, R., Grabnic, T., Sibener, S. J., and Hase, W. L. *Exploratory Direct Dynamics Simulations of O₂ Reaction with Graphene at High Temperatures*. **Journal of Physical Chemistry C**, 122, 29368–29379 (2018).
6. **Seenivasan, H.**, Jackson, B., and Tiwari, A. K. *Water Dissociation on Ni(100), Ni(110), and Ni(111): Reaction Path Approach to Mode Selectivity*. **Journal of Chemical Physics**, 146, 074705 (2017).

Invited Talks & Conference Presentations (Selected)

Invited Talks

- **Quantum Chemistry using Quantum Computers** — Inorganic and Physical Chemistry Department, **IISc Bangalore**, India (Aug 9, 2024); Department of Physics, **IIT Madras**, India (Aug 5, 2024).
- **Modelling Heterogeneous Photocatalysis using Quantum Algorithms** — **IBM Partner Forum**, IBM Studios Milano, Italy (May 13–16, 2024).

Conference Presentations

1. **Computation and Theoretical Chemists (CTC) Symposium**, University of Twente, The Netherlands (Apr 15, 2025). Contributed Lecture: “*Quantum Computing for Quantum Chemistry*.”
2. **Conference on Quantum Computing for Natural Science**, University of Copenhagen, Denmark (Apr 9–11, 2025). Poster Presentation: “*Scaling Quantum Chemistry on NISQ Hardware: Practical Insights for End Users*.”
3. **NWO Physics 2025**, Veldhoven, The Netherlands (Jan 2025). Demonstration (with Alberto Zorzato, IoP-UvA): “*A Journey of a Quantum Algorithm — From User Input through Quantum Computers to Post-Processing*.”
4. **NWO Physics 2024**, Veldhoven, The Netherlands (Jan 2024). Oral Presentation: “*New Insights into Photo-Oxidation of H₂O on TiO₂ Dimers using Quantum Algorithms*.”

5. **Lorentz Center Workshop on Spin-Enhanced Catalysis**, Leiden, The Netherlands (Nov–Dec 2023). Poster Presentation: *“Advancing Spin-Enhanced Electrocatalysis with Quantum Computing.”*
6. **NWO CHAINS 2022**, Veldhoven, The Netherlands (Dec 2022). Oral Presentation: *“Exploring Water Dissociation on Oxides using Quantum Algorithms.”*

Outreach & Collaboration

- **Ethics & Policy in Quantum Technologies:** Contributed to a white paper on ethical and legal frameworks for equitable access to quantum computing resources as part of the **EuroHPC initiative** selecting SURF to host a quantum computer in the Netherlands. Invited based on extensive experience with diverse quantum hardware platforms. *Reference: Ariana Torres Knoop, Lead Quantum Computing, SURF, The Netherlands.*
- **National Quantum Ecosystem:** Regular participant in **Quantum Delta NL** initiatives, supporting the Netherlands' national quantum technology roadmap and academic–industry partnerships. *Reference: Diederick Croese, Director of the Centre for Quantum and Society, QuantumDeltaNL.*
- **Global Capacity Building:** Mentored two proposals submitted to the **Open Quantum Initiative (CERN)**, focused on applying quantum algorithms toward **UN Sustainable Development Goals (SDGs)** with case studies from underrepresented regions in quantum research. Invited expert on quantum technologies for energy research at the **QDNL–UNICC Conference** (The Hague, 2024) and **GESDA Summit** (CERN, Geneva, 2025). *Reference: Dr. Martin Gastal, Capacity Building and Partnerships Lead, OQI – CERN.*
- **Academic & Interdisciplinary Collaboration:** Active in outreach efforts fostering collaboration across the **Van 't Hoff Institute for Molecular Sciences (UvA)**, the **Theoretical Chemistry group at Leiden Institute of Chemistry**, and the **Computer Science Department at Vrije Universiteit Amsterdam**, including supervision of a Master's project integrating quantum algorithms and hardware-aware optimization.

Honors & Certifications

- **IBM Certified Associate Developer – Quantum Computation using Qiskit v0.2X** (2023)
- **Winner, ASPEN Oss Best F&M Chemistry Poster Prize**, CHAINS Conference (Dec 2020)
- **Seal of Excellence**, Horizon Europe for proposals *PHONONSINACTION* (Mar 2019) and *SURFACEPHONON-DAMPING* (Mar 2018)
- **CSIR–Senior Research Fellowship (SRF)**, Council of Scientific and Industrial Research, Government of India (Apr 2013 – Mar 2016)
- **International Travel Grant**, IISER Kolkata, for poster presentation at the *Gordon Research Conference on Dynamics at Surfaces*, Rhode Island, USA (Aug 2015)
- **International Travel Support**, Department of Science and Technology (India), for participation in the FHI Berlin workshop, Norderney, Germany (Jul 2013)

Leadership & Mentoring

- Co-designed and co-taught MSc course *Near-Term Quantum Computing* at University of Amsterdam; secured student hardware access (IBM, IonQ, neutral atoms).
- Daily supervisor for 12x BSc/MSc/High-School students on quantum simulation, routing, and catalysis.
- Led theory section of Roadmap paper on Spin-Enhanced Catalysis (submitted to **Adv. Energy Mater.**)

References

Available upon request.